DeepNeoAG Learning Through Controlled Experiments

CS535
Nguyễn Thanh Trọng

Võ Trần Vũ Khoa
Mai Nhật Minh

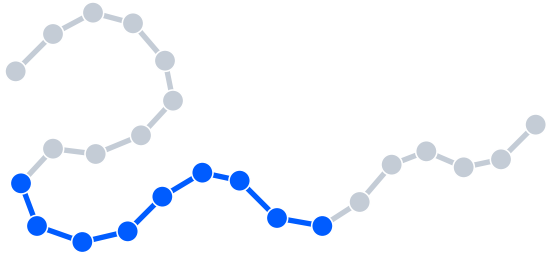
Introduction

Protein, peptide and neoantigen

A tumor mutation can change a peptide and create a new immune target

1 PROTEIN

Normal protein



processing

2 PEPTIDE

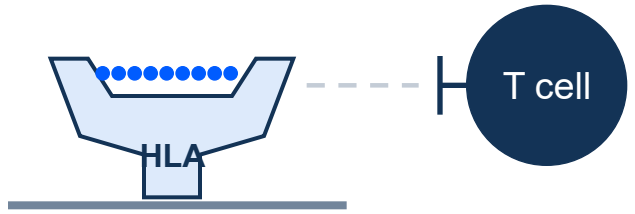
Normal peptide



Unaltered sequence

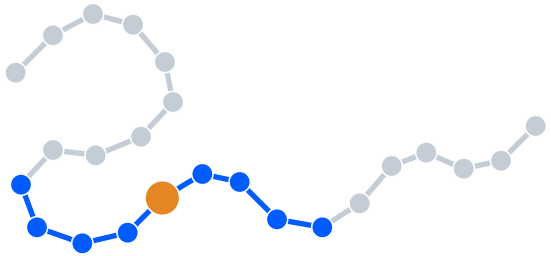
presentation

3 IMMUNE RECOGNITION



Usually tolerated as self

Mutated tumor protein



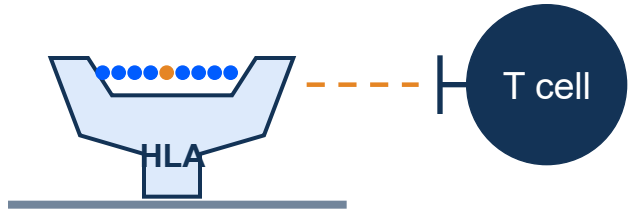
processing

Mutation-derived peptide



Candidate neoantigen

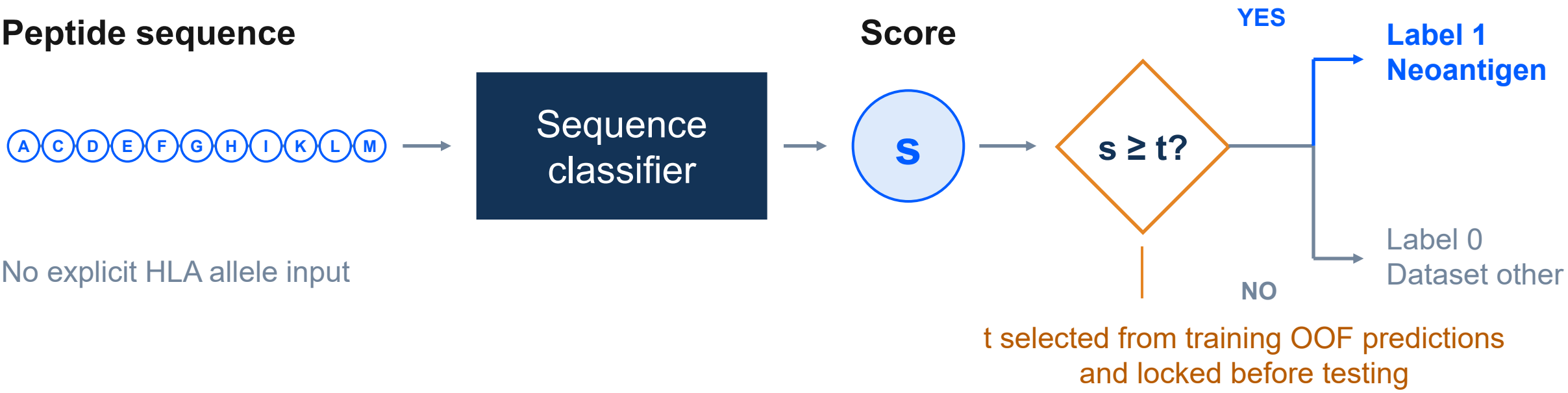
presentation



Possible T-cell recognition

Defining the prediction problem

A ranking score becomes a binary decision at a locked threshold



No explicit HLA allele input

Five questions behind the model sequence

Each approach is trained independently

Approach	Research question
Composition + LR	Are residue frequencies sufficient?
One-hot + MCNN	What can ordered learning from labels achieve?
ESM pooled LR	Do frozen pretrained representations add signal?
ESM CNN	Does local extraction help contextual embeddings?
ESM MCNN	Do multiple local scales help, with added capacity?

Methodology

Dataset provenance and published splits

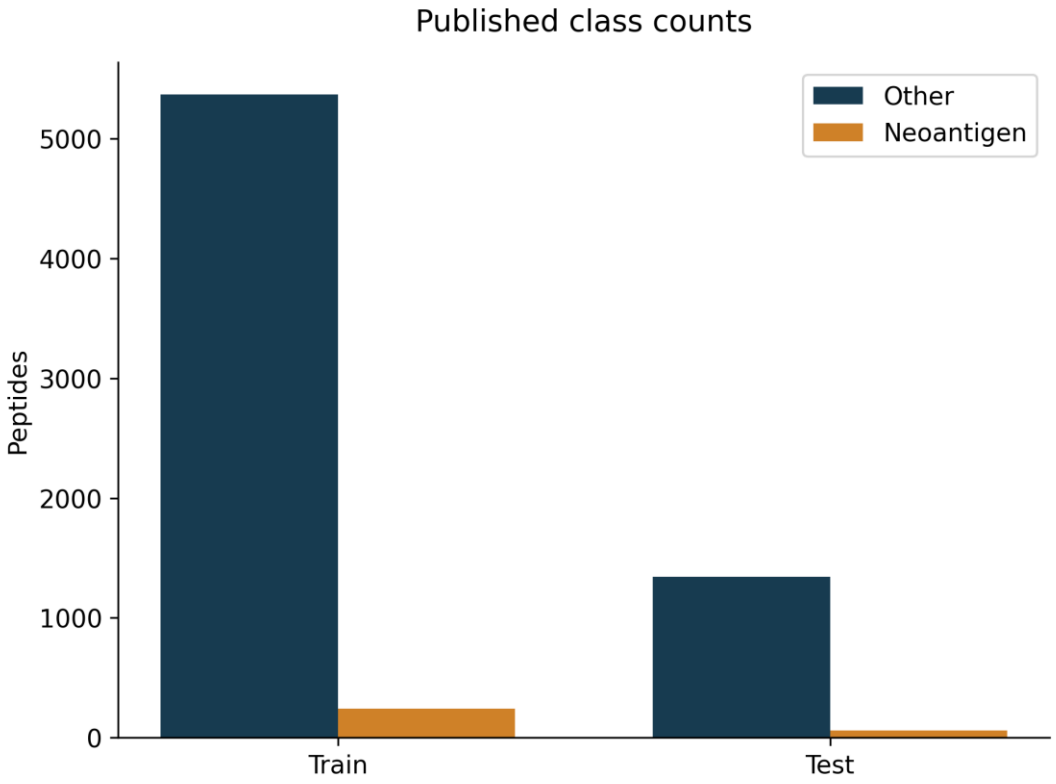
The independent test keeps its published boundary

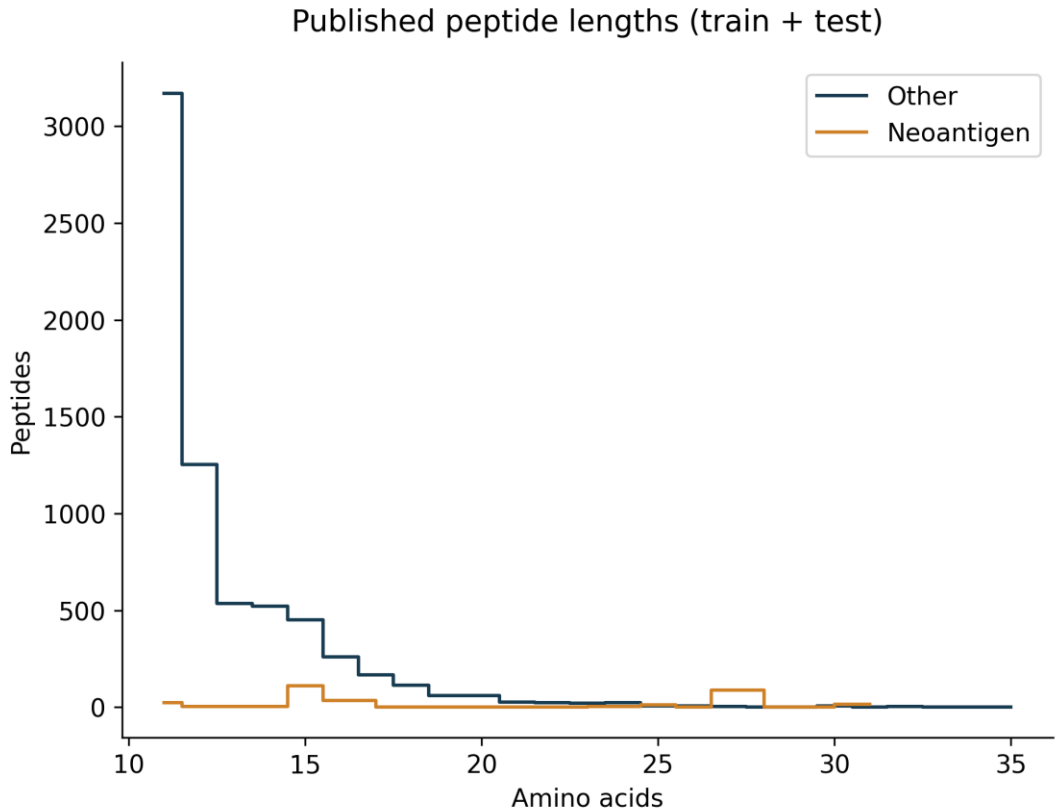


Partition	Neoantigen (1)	Other (0)	Total
Training	241	5,367	**5,608**
Independent test	**61**	1,343	**1,404**
Total	302	6,710	7,012

Understanding the data before training

Rare positives and a length-35 boundary case affect the method





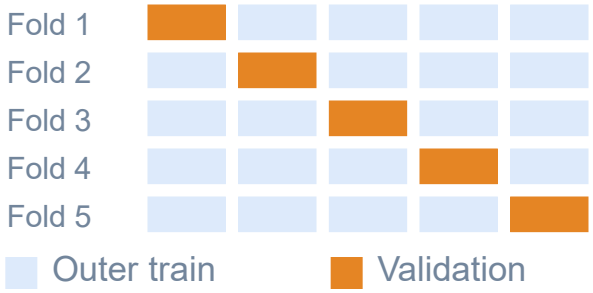
Designing a fair evaluation

Fitting, early stopping, model selection and testing use distinct roles

Training only: 5,608 records

1 CROSS-VALIDATION

Shared stratified folds



Neural: split outer-training portion



Independent test
1,404 records

OOF scores

2 SELECT AND LOCK

Configuration

Highest mean CV AP

Threshold t

Highest OOF MCC

Locked before testing

Held out until final evaluation

3 FINAL REFIT

All 5,608

training records

Neural seeds: 535, 536, 537

Locked
models + t

Final evaluation

Metric definitions and threshold policy

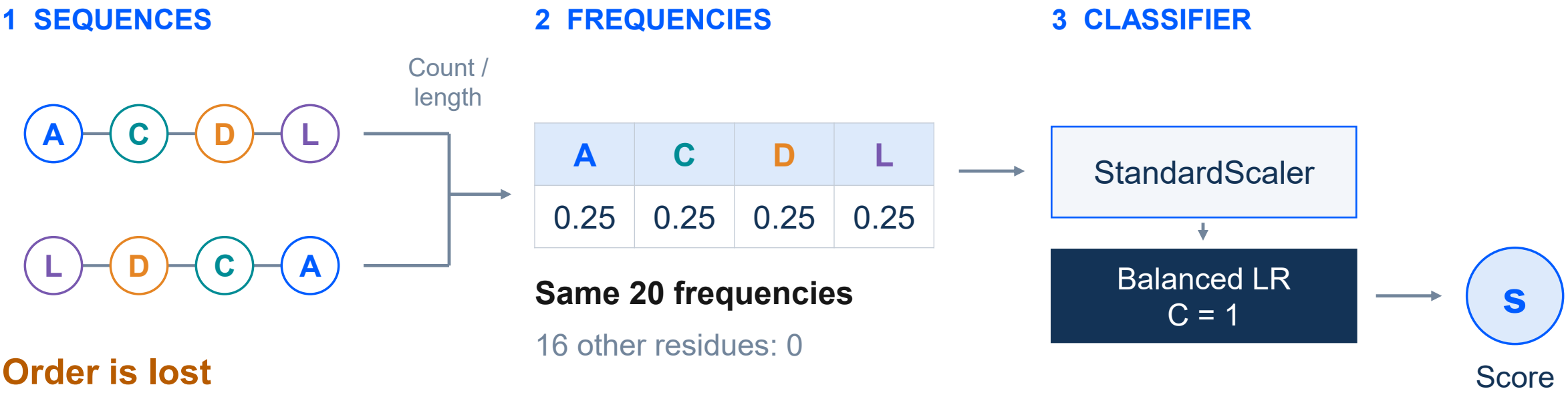
ranking quality and a binary operating point answer different questions

Measure	Meaning
Average precision (AP)	Summarizes precision across recall changes in a ranked list
ROC-AUC	Ranking discrimination across false-positive-rate levels
Precision	$TP / (TP + FP)$
Recall / sensitivity	$TP / (TP + FN)$
Specificity	$TN / (TN + FP)$
MCC	$(TP \times TN - FP \times FN) / \sqrt{[(TP+FP)(TP+FN)(TN+FP)(TN+FN)]}$

Configuration: maximize mean CV AP. Threshold: maximize MCC from seed-averaged training OOF predictions.
Independent test uses both 0.5 and the locked OOF threshold. Main slides use the OOF threshold.

Baseline 1: amino-acid composition

Composition contains some signal, but loses residue order



Order is lost

Mean CV AP	Test AP	Test MCC	OOF threshold
0.1287	**0.0714**	0.0515	0.6508

Baseline 2: one-hot MCNN

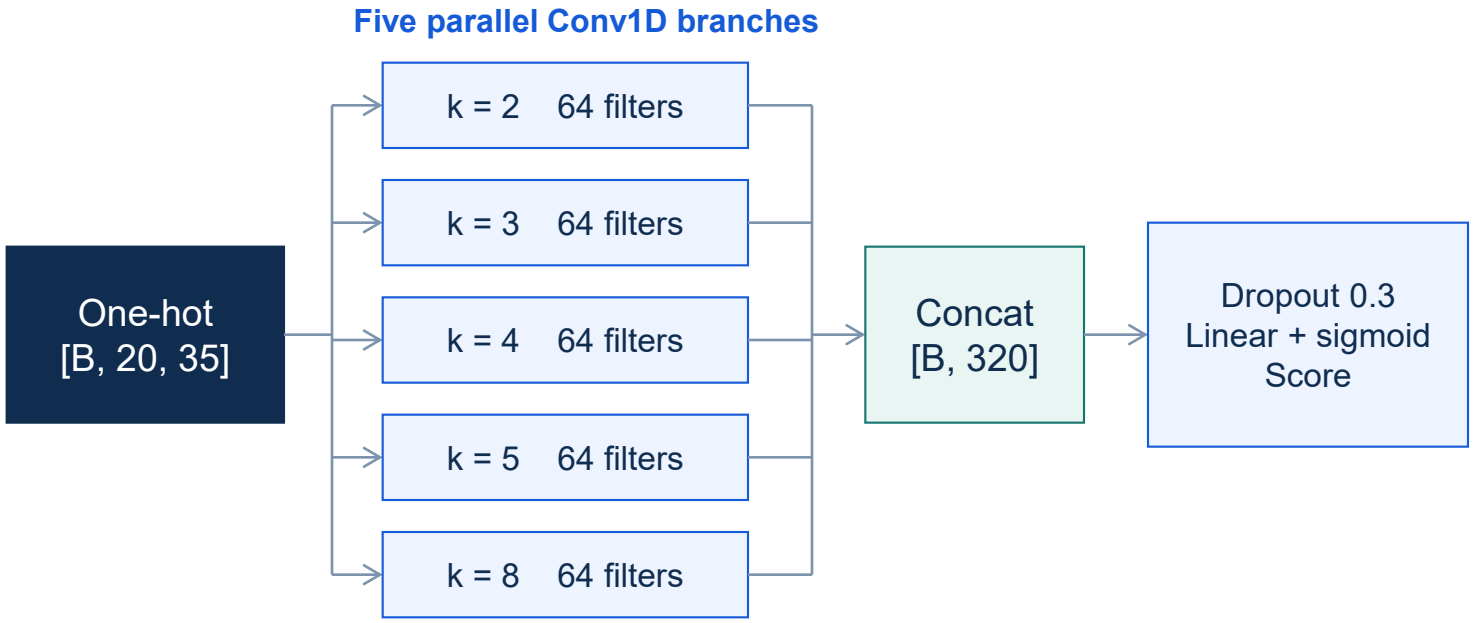
Ordered local patterns learned directly from task labels

No pretrained encoder

Five parallel windows
k = 2, 3, 4, 5, 8

64 filters per branch
320 concatenated features

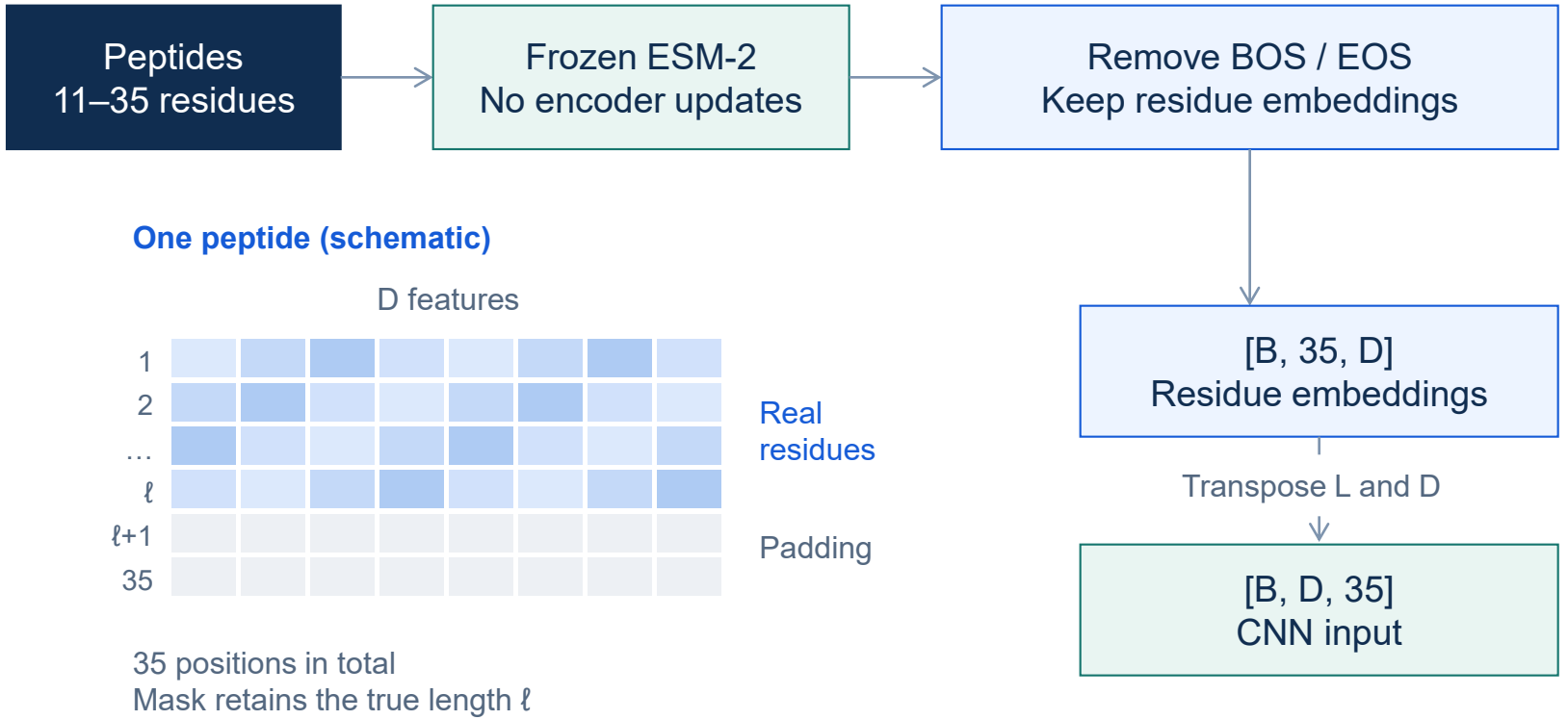
Mask padding before max pooling



Trainable parameters	CV AP	Test AP	Test MCC
28,801	0.4943	**0.5408**	0.5495

From peptides to contextual embeddings

A frozen ESM encoder gives every real residue a contextual representation



B
Peptides per batch

L
Padded residue positions

D
Features per residue

8M: D = 320
35M: D = 480

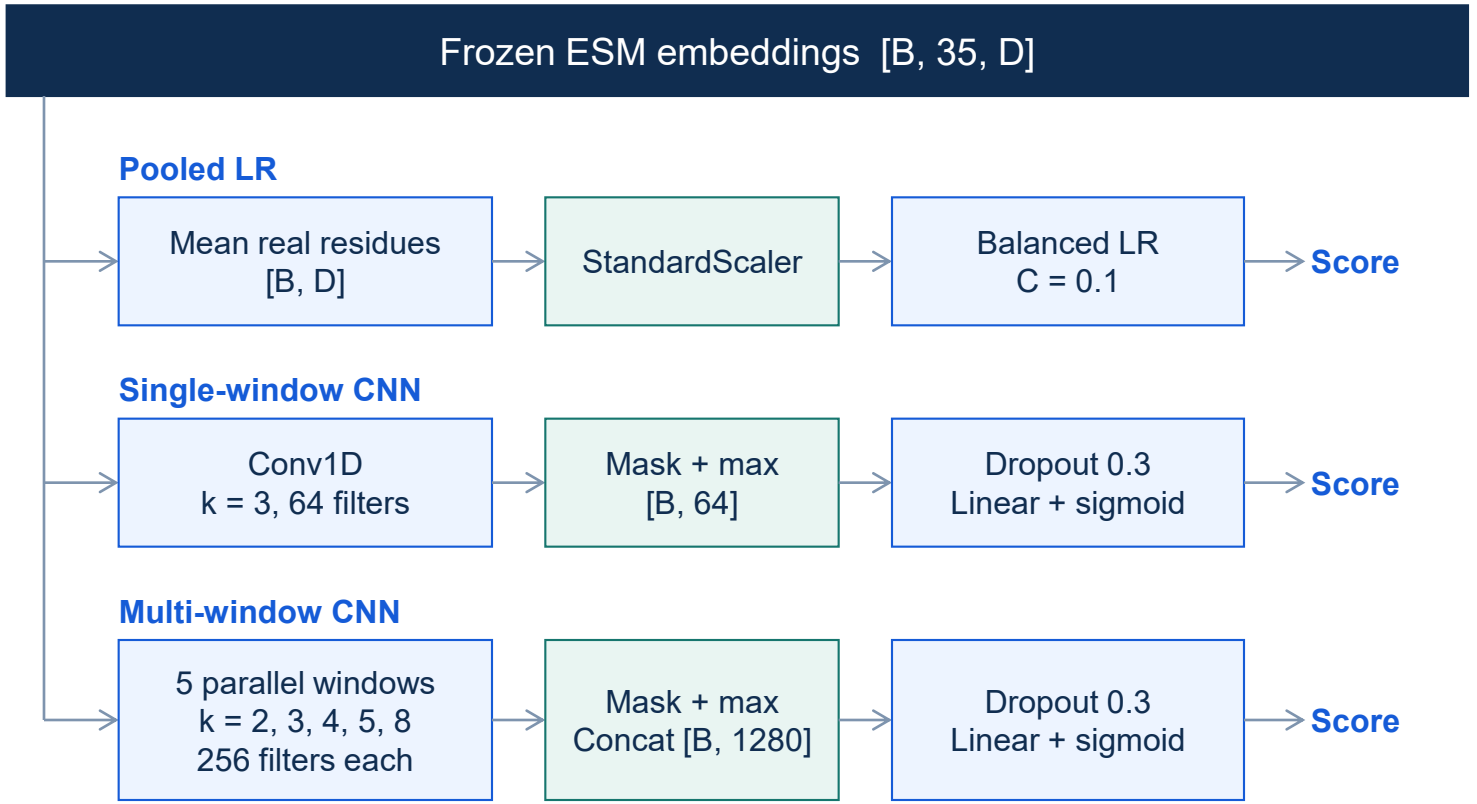
Pooling, single-window CNN and MCNN

The same frozen embeddings support three independent classifiers

Pooled LR
Real-residue mean; C = 0.1
CV / test AP: 0.3976 / 0.4764

Single-window CNN
k = 3; 64 filters
CV / test AP: 0.4868 / 0.5122

Multi-window CNN
256 filters × 5; concat 1,280
CV / test AP: 0.5142 / 0.5437



Our controlled experiment matrix

A staged, one-factor search, with 22 configurations

Changed component	Values	Held fixed / caveat
LR regularization	C = 0.1, 1, 10	Balanced LR; shared folds
Single CNN window, 8M	k = 2, 3, 4, 5, 8	64 filters; weighted loss
MCNN filters, 8M	32, 64, 128, 256 per branch	Same five windows; weighted loss
MCNN loss, 8M	Weighted / unweighted	64 filters; same windows
Frozen encoder	8M / 35M	Transfer selected CNN settings; D and head size change

Selected configurations

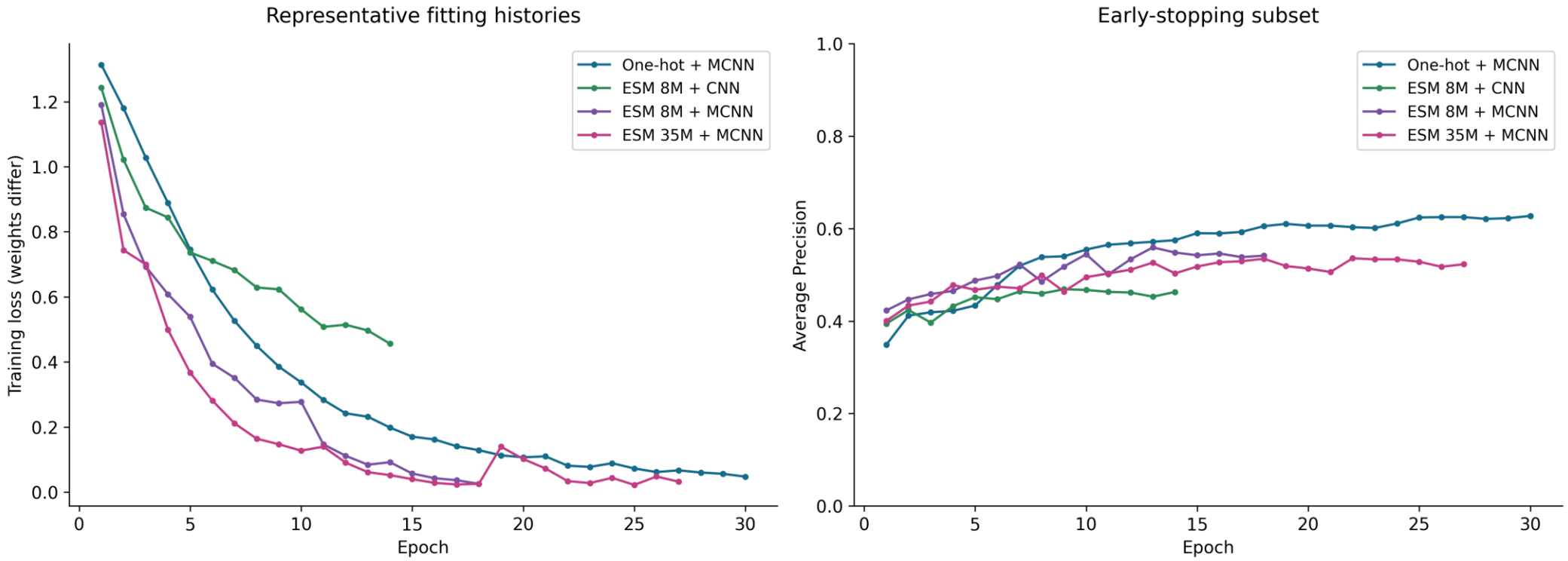
settings locked before independent testing

Model	Selected setting	Parameters	OOF t	Refit epochs
Composition + LR	C = 1	21	0.650830	Not applicable
One-hot + MCNN	k=2,3,4,5,8; 64 filters	**28,801**	0.512465	23
ESM 8M + pooled LR	C = 0.1	321	0.977457	Not applicable
ESM 8M + CNN	k=3; 64 filters	61,569	0.930151	13
ESM 8M + MCNN	**k=2,3,4,5,8; 256 filters**	1,804,801	0.766193	16
ESM 35M + pooled LR	C = 0.1	481	0.945360	Not applicable
ESM 35M + CNN	k=3; 64 filters	92,289	0.732428	10
ESM 35M + MCNN	**k=2,3,4,5,8; 256 filters**	**2,705,921**	0.632288	14

Results

Tracking learning and computational cost

Early-stop AP can plateau while fitting loss continues to decline



Extraction (8M / 35M)

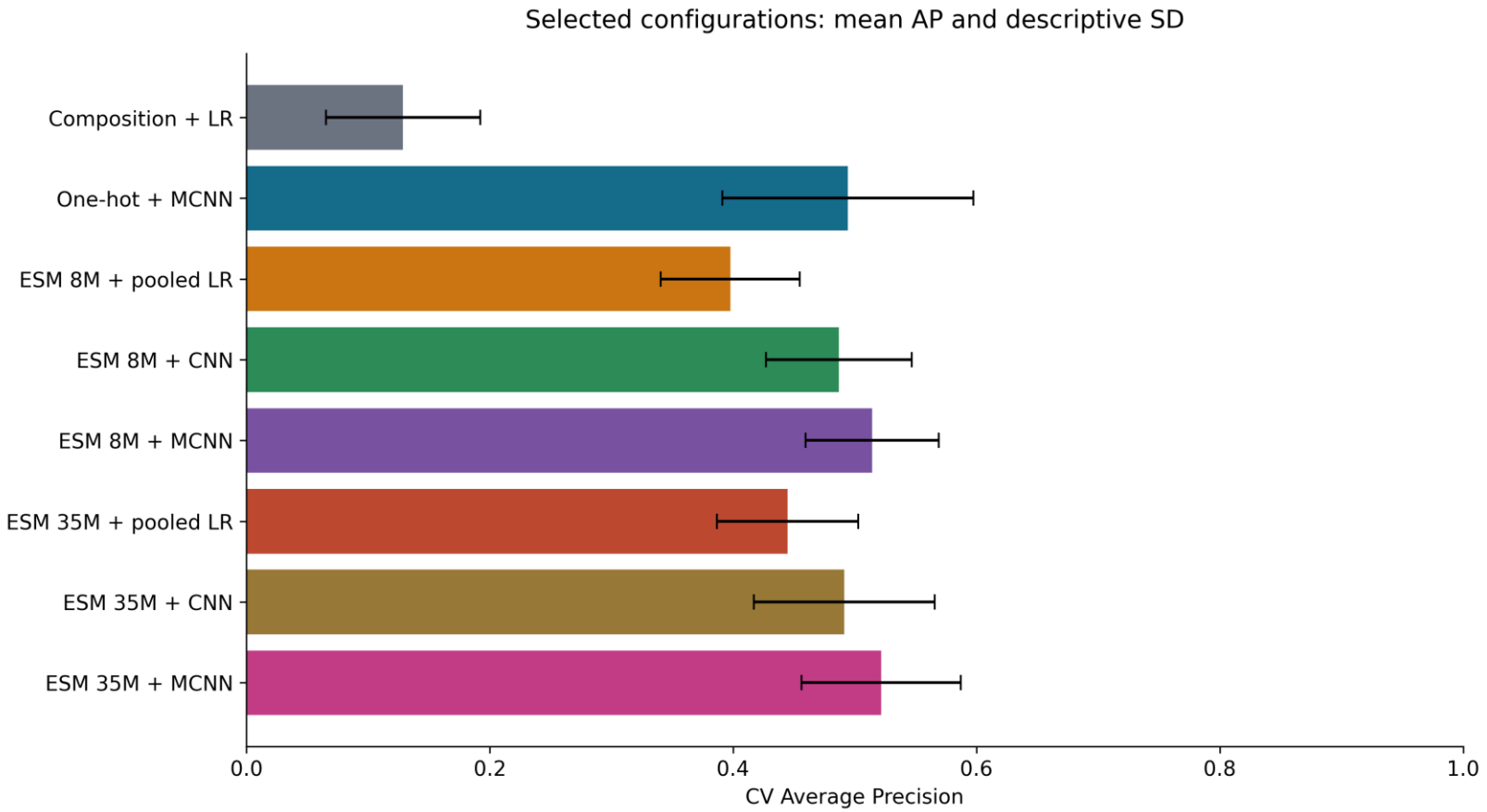
9.53 s / 13.21 s

Mean MCNN CV fit (one-hot / 8M / 35M)

10.26 s / 11.50 s / 13.52 s

What we observed across folds and seeds

Mean AP and descriptive spread answer different questions



Mean CV AP ± SD

One-hot MCNN
0.4943 ± 0.1032

8M MCNN
0.5142 ± 0.0547

35M MCNN
0.5216 ± 0.0655

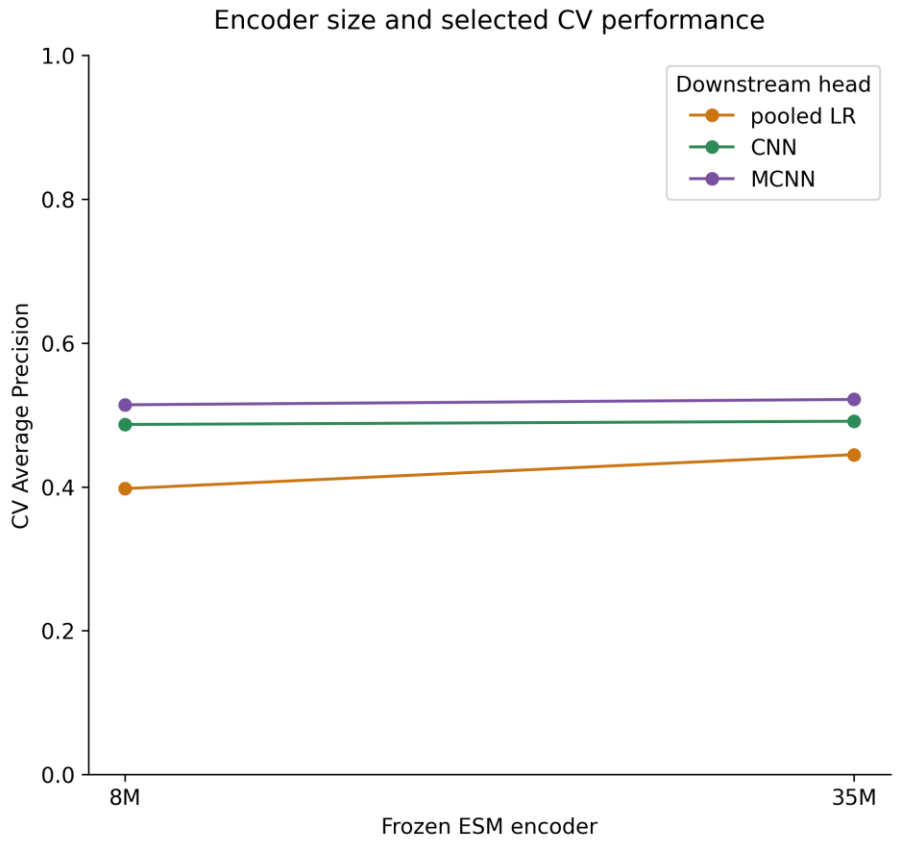
Fixed defaults versus tuned selections

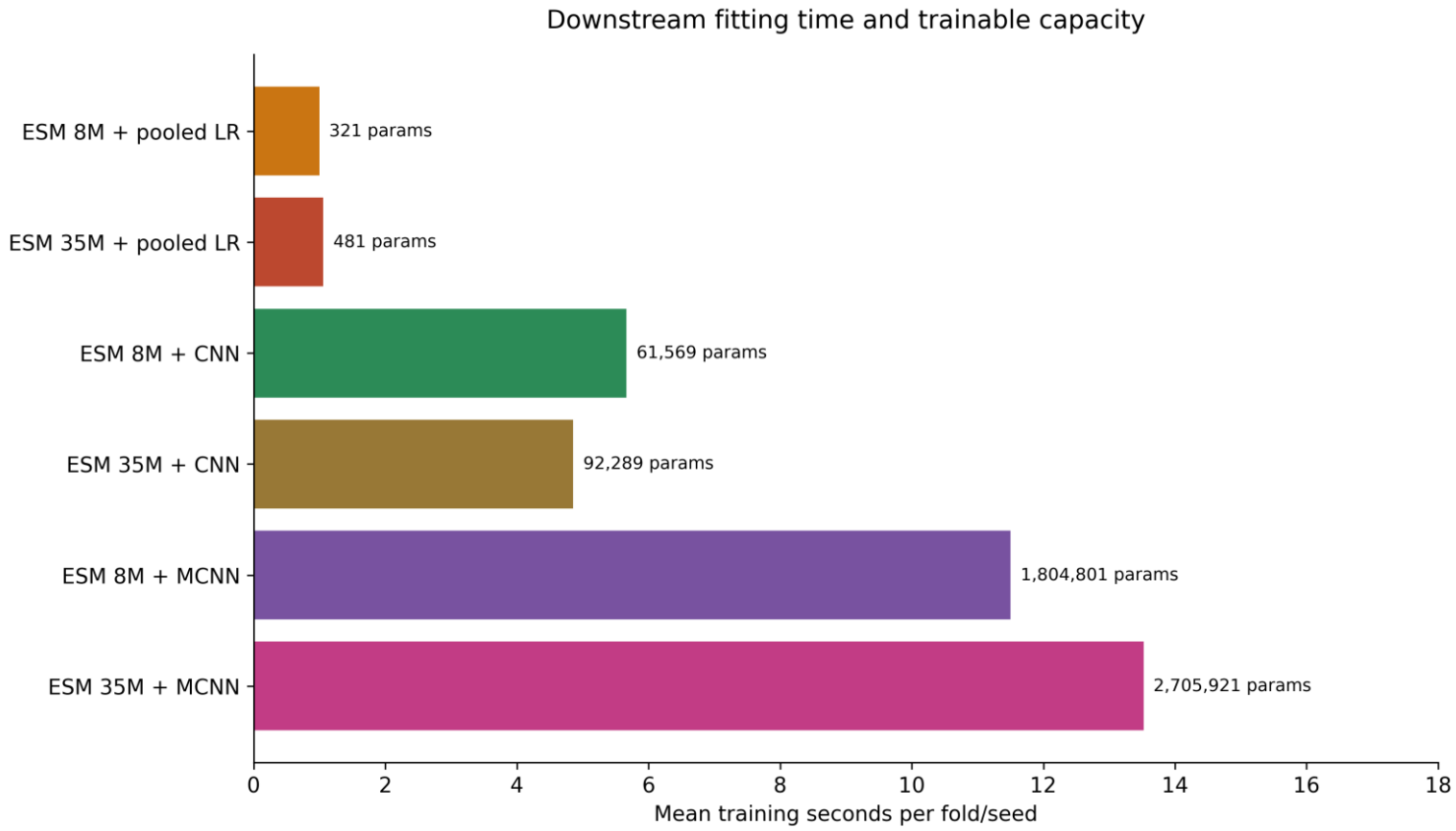
comparison includes the model-selection step

Model	Default CV AP	Selected CV AP	Default params	Selected params
Composition + LR	0.1287	0.1287	21	21
One-hot + MCNN	0.4943	0.4943	28,801	28,801
ESM 8M + pooled LR	0.3264	**0.3976**	321	321
ESM 8M + CNN	0.4868	0.4868	61,569	61,569
ESM 8M + MCNN	0.5013	**0.5142**	451,201	**1,804,801**

Encoder size, head quality and cost

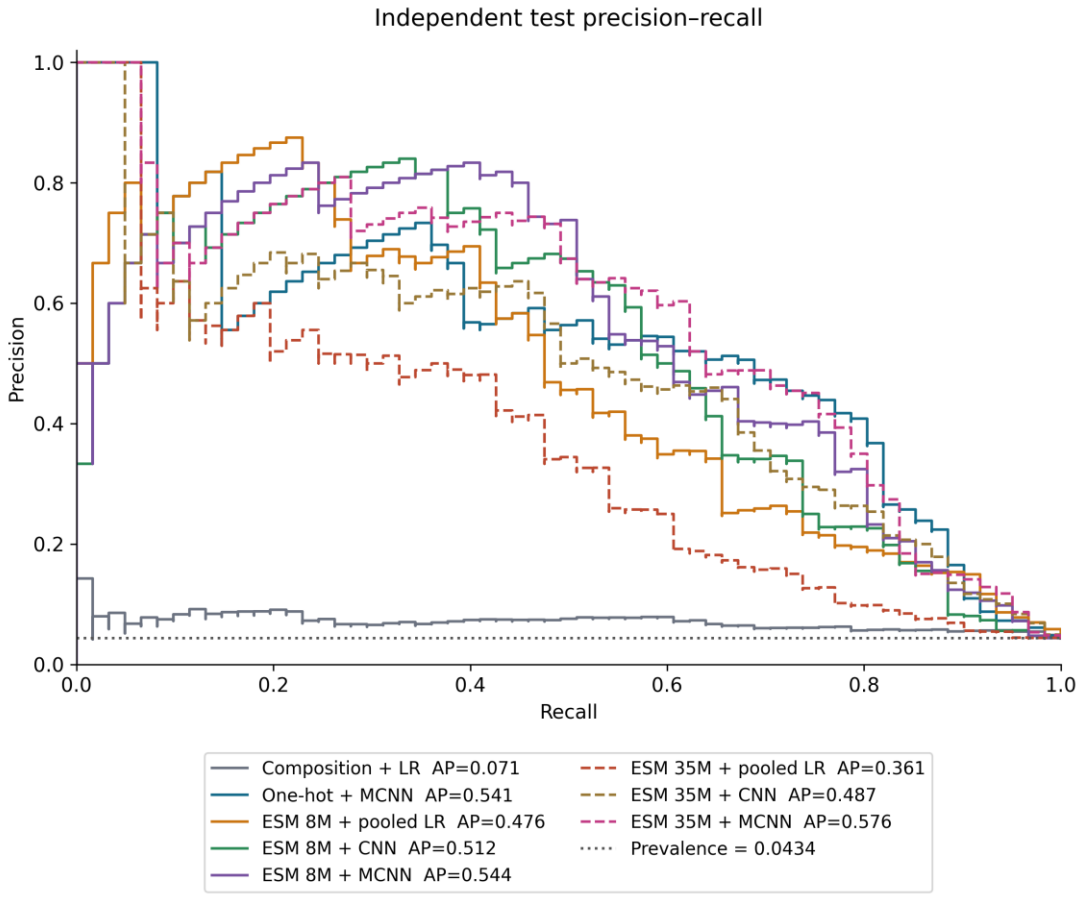
transferred CNN settings, with changed D and head parameter count





Independent test: a final check

ESM 35M MCNN has the highest observed AP, with uncertain differences



Locked model	AP	MCC
One-hot MCNN	0.5408	0.5495
8M MCNN	0.5437	0.5512
35M MCNN	**0.5758**	**0.5513**

35M MCNN: 44 / 61 positives found
52 false positives

AP 95% CI: [0.4631, 0.7061]

Paired AP difference vs one-hot
95% CI: [−0.0444, 0.1151]

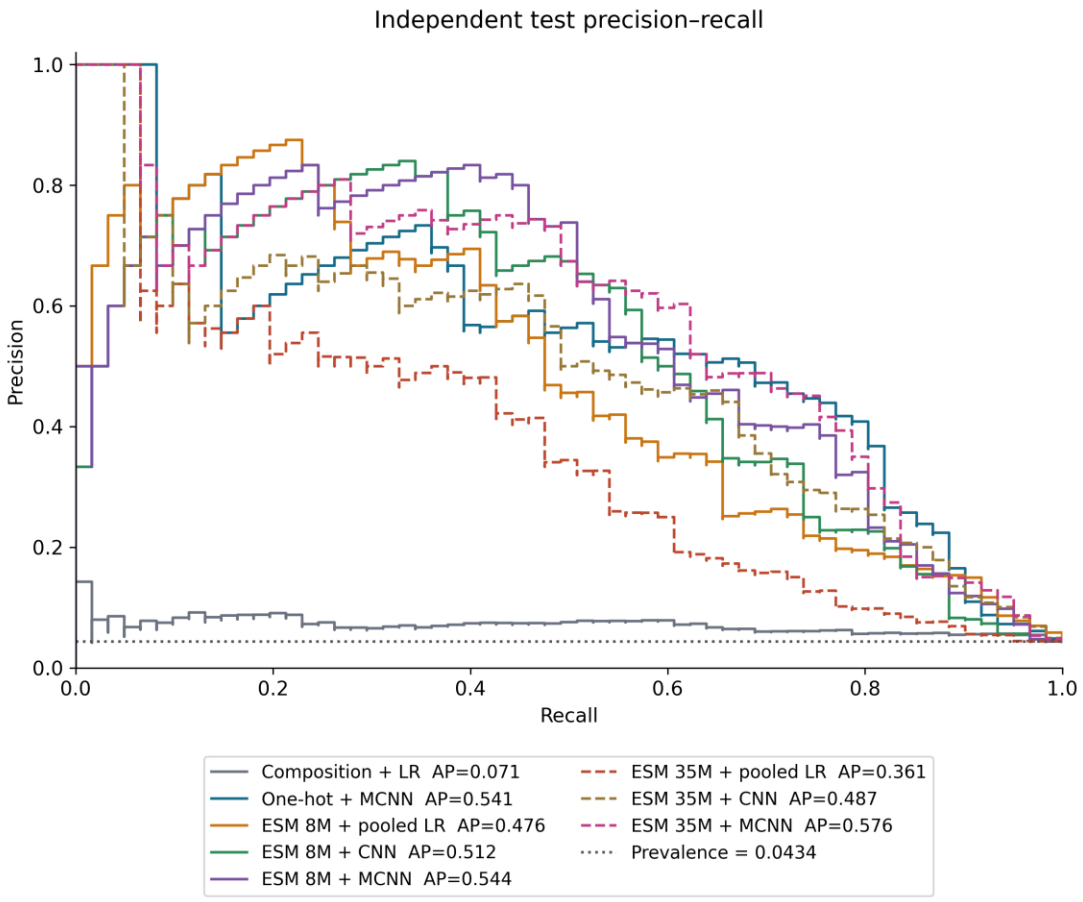
Complete independent-test metrics

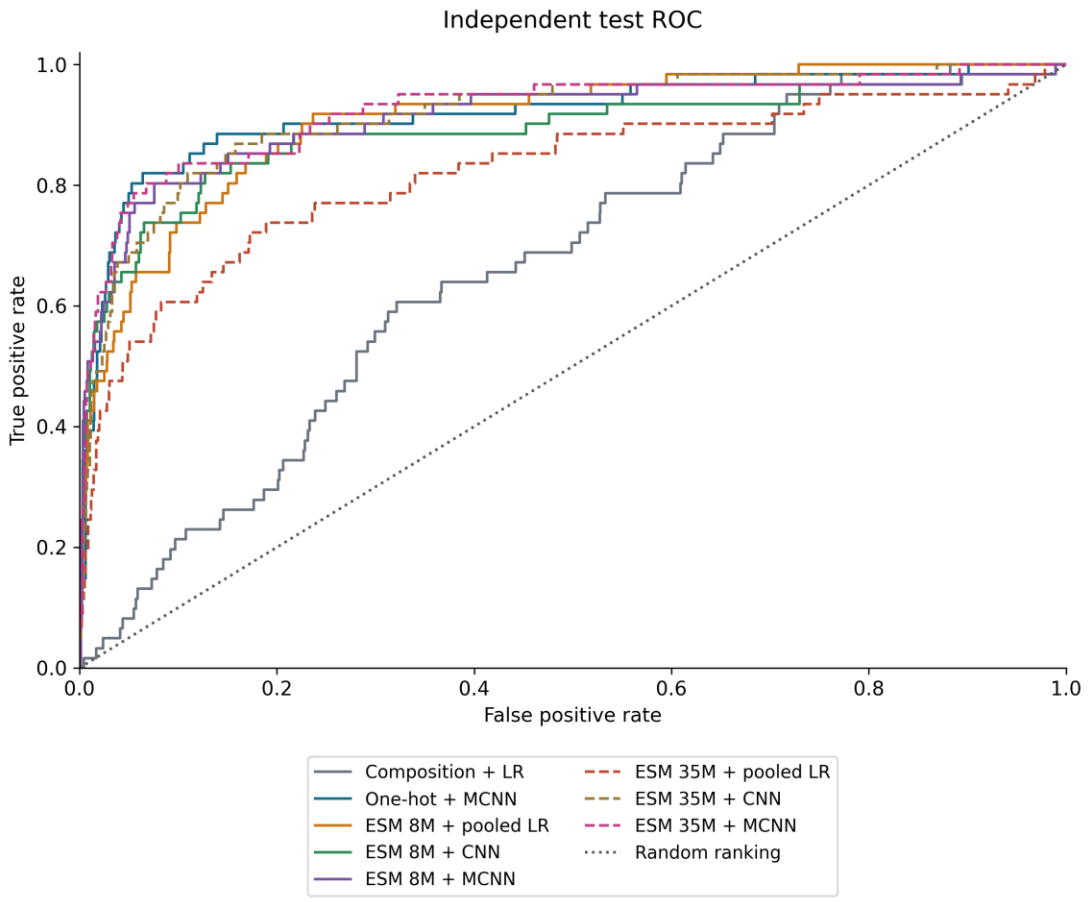
ensemble probabilities and locked OOF thresholds

Model	AP	ROC-AUC	MCC	Precision	Recall	Specificity	TP / FP
Composition + LR	0.0714	0.6625	0.0515	0.0690	0.2295	0.8593	14 / 189
One-hot + MCNN	0.5408	0.9179	0.5495	0.5362	0.6066	0.9762	37 / 32
ESM 8M + pooled LR	0.4764	0.9077	0.4743	0.6667	0.3607	0.9918	22 / 11
ESM 8M + CNN	0.5122	0.8880	0.5262	**0.7143**	0.4098	**0.9926**	25 / 10
ESM 8M + MCNN	0.5437	0.9108	0.5512	0.6000	0.5410	0.9836	33 / 22
ESM 35M + pooled LR	0.3614	0.8213	0.3693	0.5143	0.2951	0.9873	18 / 17
ESM 35M + CNN	0.4870	0.9154	0.4938	0.4557	0.5902	0.9680	36 / 43
ESM 35M + MCNN	**0.5758**	**0.9224**	**0.5513**	0.4583	**0.7213**	0.9613	**44 / 52**

Complete test precision–recall and ROC

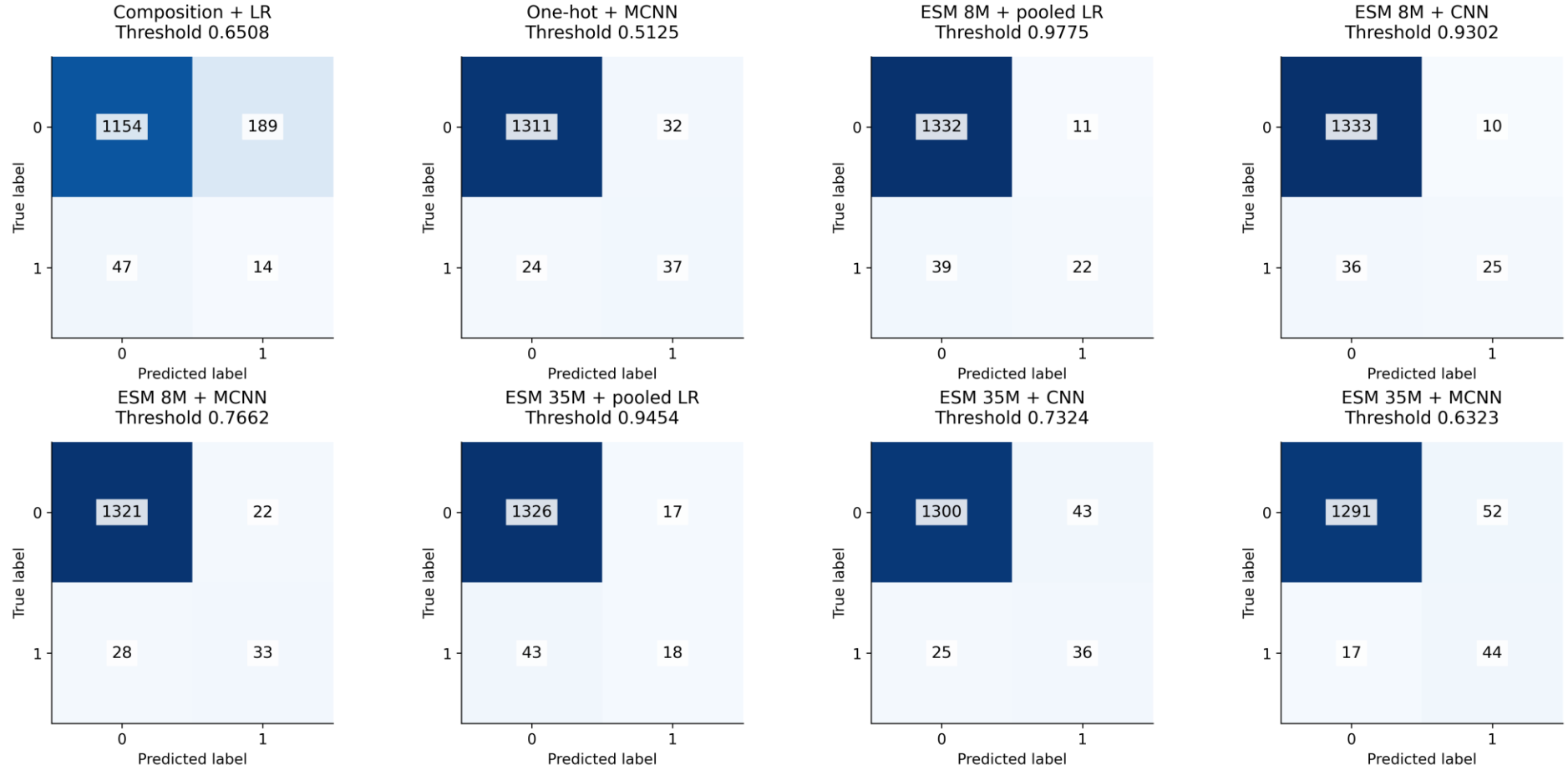
all eight locked methods on the same independent test





All eight confusion matrices

raw counts at locked OOF-selected thresholds



Conclusion

Answers to the five research questions

Five questions from the introduction, answered with independent-test evidence

Question	Independent-test answer
Are residue frequencies sufficient?	Limited signal here: composition + LR AP 0.0714.
What can ordered learning from labels achieve?	One-hot MCNN: AP 0.5408 with 28,801 trainable parameters.
Do frozen pretrained representations add signal?	Yes: 8M pooled LR AP 0.4764. Below one-hot MCNN in this test.
Does local extraction help contextual embeddings?	8M CNN AP 0.5122 vs pooled LR 0.4764. The classifier also changes.
Do multiple local scales help, with added capacity?	8M MCNN AP 0.5437 vs CNN 0.5122. More windows and head capacity.